HOCHSCHULE RAVENSBURG-WEINGARTEN UNIVERSITY OF APPLIED SCIENCES

GPIO Output Stages in Simulation

Push-pull, open drain and open collector: switching speed, voltage and current limits

Embedded Computing – Master of Engineering – Worksheet

Overview

Every GPIO pin ends in an output stage: two transistors (push-pull) or one transistor to ground (open drain, open collector). In this worksheet you will take these stages apart in a circuit simulator. You will find out how much current a pin can deliver, how fast it can switch a load, why a pull-up resistor makes a bus slow, and which voltages and currents destroy a pin or the circuit behind it.

Learning outcomes. After this worksheet you can

- model a GPIO output stage by on-resistance and saturation current and relate the model to the V_{OH}/V_{OL} specifications of a data sheet,
- predict rise and fall times, the maximum toggle frequency and the dynamic power of a push-pull output,
- dimension the pull-up resistor of an open-drain bus such as I^2C from rise-time and V_{OL} limits,
- judge where the maximum output voltage and current of a pin come from (body diodes, injection current, back-powering, transistor breakdown and saturation) and protect a driver against inductive kick-back,
- recognise when a PCB trace has to be treated as a transmission line and terminate it.

Organisation. About 4 hours, in teams of two. You do not have to calculate: all results of the model are given in tables. Tasks marked [P] (preparation) ask you to read these values and understand where they come from *before* you open the simulation. [S] means simulate, measure and enter your values in the empty column *Simulation* of the tables. [D] means discuss and explain. Hand in a short report (PDF): for each task the completed tables, screenshots of the scopes and your explanation. When a measurement differs from the model value by more than about $10\,\%$, explain why.

Tool. Falstad CircuitJS runs in the browser: `https://www.falstad.com/circuit/`. Each part of this sheet has a link that opens the prepared circuit. The circuit files are also in the folder `examples/interfaces/falstad/` of the lecture repository (open them with *File* → *Open File*).

Working with CircuitJS

- Hover over an element to see its voltages, currents and power in the lower right corner.
- Click on a logic input (H/L) or a switch to toggle it. Double-click an element to edit its value. Sliders on the right change the values marked in the circuit.
- Scopes: right-click → *Properties* sets the time base ("scope speed"), the scale and shown values (e.g. average, peak-to-peak). Drag with the mouse in a scope to measure time differences.
- *Reset* restarts the simulation, *Run/STOP* freezes it for reading values.
- *Options* → *Other Options* lets you change the simulation time step. If edges look jagged, make the time step smaller.

The transistor model used in all circuits

The MOSFETs follow the square-law (Shichman–Hodges) model with threshold $V_T = 0.6\,\text{V}$ and

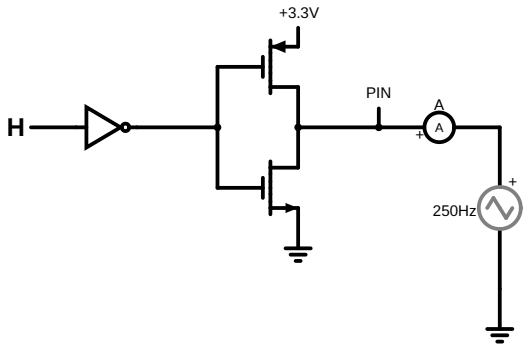
$$I_D = \begin{cases} \beta\left[(V_{GS} - V_T)\,V_{DS} - \tfrac{1}{2}V_{DS}^2\right] & V_{DS} < V_{GS} - V_T \quad \text{(triode)} \\ \tfrac{\beta}{2}\,(V_{GS} - V_T)^2 & V_{DS} \geq V_{GS} - V_T \quad \text{(saturation)} \end{cases}$$

with $\beta_N = 12\,\text{mA/V}^2$ for the NMOS and $\beta_P = 8\,\text{mA/V}^2$ for the PMOS of the output stage (PMOS: use V_{SG}, V_{SD}). Both have a body diode. The supply is $V_{DD} = 3.3\,\text{V}$. The model knows no gate or junction capacitances, no temperature and no breakdown. Pin and load capacitances are drawn as explicit capacitors. Keep in mind what the model cannot show when you interpret the results.

1 Push-pull output: static characteristic

▶ open in Falstad (`A_pushpull_characteristic.txt`)

The logic input is the bit in the output register. Two inverters (the predriver and the CMOS stage) make the pin follow it. A triangle voltage source acts as a curve tracer: it forces the pin voltage from $-0.5\,\text{V}$ to $3.8\,\text{V}$, and an ammeter measures the pin current (positive = current out of the pin). The left scope shows the current over the voltage, the right scope the power in both transistors.



Task 1.1 [P] Table 1.1 gives the on-resistance (valid for small V_{DS}) and the saturation current of both transistors for $V_{GS} = V_{DD}$. Which one is the stronger transistor, and why do chip designers make the PMOS larger than the NMOS?

Table 1.1: model values of the output transistors ($V_{GS} = 3.3\,\text{V}$)

	formula	NMOS	PMOS
β		$12\,\text{mA/V}^2$	$8\,\text{mA/V}^2$
R_{on}	$1/(\beta(V_{GS} - V_T))$	$30.9\,\Omega$	$46.3\,\Omega$
I_{sat}	$\frac{\beta}{2}(V_{GS} - V_T)^2$	$43.7\,\text{mA}$	$29.2\,\text{mA}$

Task 1.2 [S] Set the output to H and then to L. From the curve, read V_{OH} and V_{OL} at the currents of Table 1.2 (Run/STOP and hovering over the PIN label helps) and enter them next to the model values. Also read the short-circuit currents (pin at $0\,\text{V}$ for H, pin at $3.3\,\text{V}$ for L).

Table 1.2: output levels over the load current

$\lvert I \rvert$	2 mA	4 mA	8 mA	12 mA	20 mA	short circuit
V_{OH}, model V_{OH}, simulation	3.21 V	3.11 V	2.90 V	2.67 V	2.11 V	29.2 mA
V_{OL}, model V_{OL}, simulation	0.06 V	0.13 V	0.26 V	0.40 V	0.71 V	43.7 mA

Task 1.3 [D] The RP2040 data sheet specifies $V_{OH} \geq 2.62\,\text{V}$ and $V_{OL} \leq 0.5\,\text{V}$ at a load current equal to the selected drive strength (2, 4, 8 or 12 mA) and $IOVDD = 3.3\,\text{V}$. The model delivers $12.8\,\text{mA}$ at $V_{OH} = 2.62\,\text{V}$ and sinks $12\,\text{mA}$ at $V_{OL} = 0.40\,\text{V}$. Which drive strength setting does the model represent? Drive strength is implemented by switching transistor "fingers" in parallel, i.e. by scaling β ($I_D \propto \beta$): the $4\,\text{mA}$ setting needs $\beta_P = 2.5\,\text{mA/V}^2$. What fraction of the fingers is that, and what does it mean for the switching speed (part 3)?

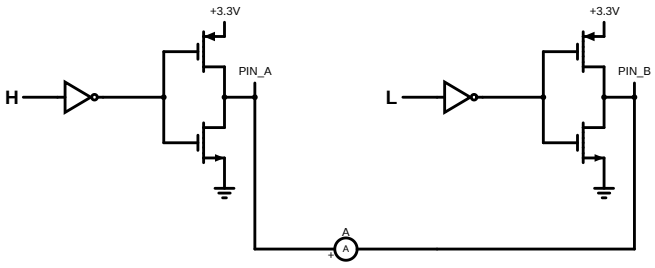
Task 1.4 [S/D] A red LED ($V_F \approx 2.0\,\text{V}$) with a $100\,\Omega$ series resistor is connected from the pin to GND. The naive calculation gives $I = (3.3\,\text{V} - 2.0\,\text{V})/100\,\Omega = 13\,\text{mA}$. Draw the load line $I = (V_{PIN} - V_F)/R$ into your measured characteristic (it passes through $2.0\,\text{V}/0\,\text{mA}$ and $3.3\,\text{V}/13\,\text{mA}$) and read the operating point. The model gives $V_{PIN} = 2.86\,\text{V}$ and $I = 8.6\,\text{mA}$. Why is the current a third lower than the naive value? Optional: verify it by replacing the sweep source with an LED and a resistor (*Draw → Outputs and Labels → Add LED*).

Task 1.5 [D] Look at the curve below $0\,\text{V}$ and above $3.3\,\text{V}$. Which element conducts there? Relate this to the absolute maximum rating "pin voltage $-0.5 \ldots IOVDD + 0.5\,\text{V}$" of the RP2040 and to the term *injection current*. Why can a large injection current upset an analogue measurement or even a digital circuit on the same chip?

Task 1.6 [S/D] Read the maximum power in the transistors from the right scope (model: $96\,\text{mW}$ in the PMOS for a short circuit to GND, $110\,\text{mW}$ at $-0.5\,\text{V}$). Many microcontrollers allow $50\,\text{mA}$ to $100\,\text{mA}$ in total for all pins of one supply domain. What does this mean for the number of pins that may drive LEDs at the same time, and what happens to the chip temperature if a pin is shorted to GND?

2 Two push-pull outputs on one line

▶ **open in Falstad** (`C_bus_contention.txt`)



Task 2.1 [P/S] Output A drives high, output B drives low. The line voltage follows from $I_{D,P,A}(V) = I_{D,N,B}(V)$; both transistors work in the triode region. Measure the values of Table 2.1 in the simulation.

Table 2.1: contention between two push-pull outputs

	line voltage	current	power PMOS A	power NMOS B
model	$1.09\,\mathrm{V}$	$28.2\,\mathrm{mA}$	$62\,\mathrm{mW}$	$31\,\mathrm{mW}$
simulation				

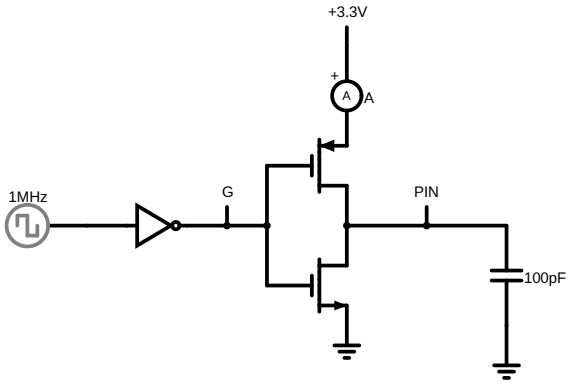
Task 2.2 [D] What logic level does a receiver see on the line?

Task 2.3 [D] Name three situations in embedded practice where this happens. How do tri-state outputs, open-drain outputs and series resistors prevent the damage? (A series resistor of about $290\,\Omega$ in each output limits the contention current to 5 mA: $2R + R_{on,P} + R_{on,N} = 3.3\,\mathrm{V}/5\,\mathrm{mA}$. What does it cost?)

3 Push-pull output: switching speed and dynamic power

▶ open in Falstad (`B_pushpull_capacitive_load.txt`)

The square wave is the program toggling the pin. The predriver is a logic inverter with an adjustable slew rate (how fast the gate voltage changes). The output drives the load capacitance C_L (pin, trace, inputs of other chips). An ammeter measures the current taken from the supply.



Task 3.1 [P] Table 3.1 gives the rise and fall times (10 % to 90 %) from two models: (a) the RC approximation $t_r = 2.2\,R_{on}C_L$, (b) the square-law model, in which the transistor starts in saturation (constant current I_{sat}) and enters the triode region when $V_{DS} = V_{GS} - V_T$ (integration of $dt = C_L\,dV/I_D(V)$). Why is the RC approximation too optimistic?

Task 3.2 [S] Set the predriver slew rate to its maximum and measure t_r and t_f for the three capacitances (set the scope time base accordingly, or zoom in with Run/STOP). For which C_L does the predriver dominate the edge?

Table 3.1: rise and fall times at the pin

	$C_L = 10$ pF		$C_L = 100$ pF		$C_L = 1$ nF	
	t_r	t_f	t_r	t_f	t_r	t_f
(a) RC model	1.0 ns	0.68 ns	10.2 ns	6.8 ns	102 ns	68 ns
(b) square-law model	1.36 ns	0.90 ns	13.6 ns	9.0 ns	136 ns	90 ns
simulation						

Task 3.3 [S/D] Increase the toggle frequency with $C_L = 1$ nF until the pin no longer reaches valid logic levels ($V_{IH} = 0.7\,V_{DD}$, $V_{IL} = 0.3\,V_{DD}$). Compare with the rule of thumb $f_{max} \approx 1/(2\,t_r) \approx 3.7$ MHz (slower edge). Why do data sheets specify the maximum GPIO frequency only together with a load capacitance?

Task 3.4 [S] Set $f = 1$ MHz and the fastest slew rate. Read the average supply current (right scope) for the three capacitances and compare with $I = C_L V_{DD} f$. The energy taken from the supply per cycle is $C_L V_{DD}^2$, independent of R_{on}. Where is it dissipated?

Table 3.2: average supply current at 1 MHz

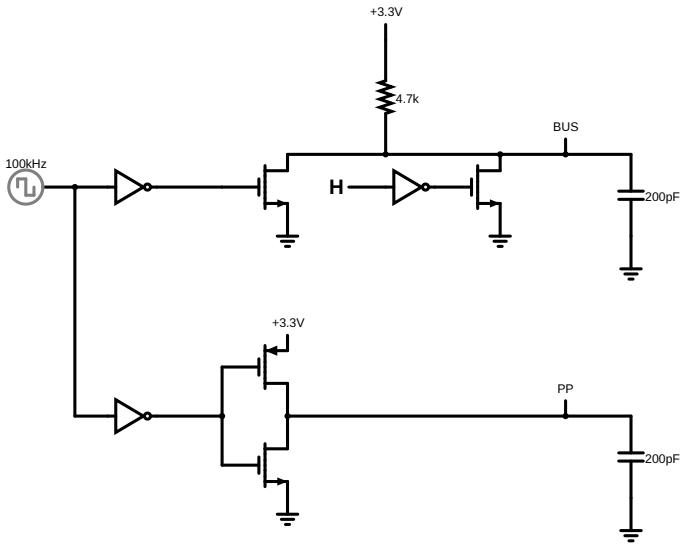
	$C_L = 10$ pF	$C_L = 100$ pF	$C_L = 1$ nF
$C_L V_{DD} f$	33 µA	330 µA	3.3 mA
simulation			

Task 3.5 [S/D] Set $C_L = 1$ pF and reduce the predriver slew rate to $0.02\,\mathrm{V/ns}$. Observe the supply current. Both transistors conduct while $V_T < V_G < V_{DD} - V_T$ (0.6 V to 2.7 V), at $0.02\,\mathrm{V/ns}$ that is 105 ns per edge. Measure the average supply current and compare it with $C_L V_{DD} f = 3.3$ µA (simulation: _____). Why do microcontroller input pins have Schmitt triggers, and why does a slowly changing analogue voltage on a digital input increase the current consumption?

4 Open-drain bus versus push-pull

► open in Falstad (`D_open_drain_bus.txt`)

Two open-drain devices share a line with one pull-up resistor R_P, as on an I²C bus. Device 1 sends a clock; device 2 can hold the line low. A push-pull stage drives the same capacitance for comparison.



Task 4.1 [P] The I²C rise time is measured from 30 % to 70 % of V_{DD}. For the RC charge $V(t) = V_{DD}(1 - e^{-t/\tau})$ it is $t_r = \tau \ln(0.7/0.3) = 0.8473\, R_P C_b$, i.e. 796 ns for $R_P = 4.7$ kΩ and $C_b = 200$ pF.

Task 4.2 [S] Measure the rise and fall times of BUS and the rise time of PP (same thresholds). Explain the difference between rising and falling edge of the open-drain line.

Table 4.1: edges with $R_P = 4.7\,\mathrm{k\Omega}$, $C_b = 200$ pF (30 % to 70 %)

	t_r BUS	t_f BUS	t_r PP
model (RC)	$0.8473\, R_P C_b = 796$ ns	$0.8473\, R_{on,N} C_b \approx 5$ ns	$0.8473\, R_{on,P} C_b \approx 8$ ns
simulation			

Task 4.3 [P] Table 4.2 gives the permitted range of R_P for $C_b = 200$ pF and $V_{DD} = 3.3$ V according to the I²C specification (NXP UM10204): $V_{OL} \le 0.4$ V at a sink current of 3 mA, $t_r \le 1000$ ns (Standard-mode, 100 kHz) and $t_r \le 300$ ns (Fast-mode, 400 kHz). Which limit comes from which requirement?

Table 4.2: pull-up range for $C_b = 200$ pF

	$R_{P,min} = \frac{V_{DD} - V_{OL}}{I_{OL}}$	$R_{P,max} = \frac{t_r}{0.8473\,C_b}$	standard values
Standard-mode	$967\,\Omega$	$5.9\,\text{k}\Omega$	$1\,\text{k}\Omega$...$5.6\,\text{k}\Omega$
Fast-mode	$967\,\Omega$	$1.77\,\text{k}\Omega$	$1\,\text{k}\Omega$...$1.5\,\text{k}\Omega$

Task 4.4 [S] Check the limits in the simulation: complete Table 4.3. Then increase C_b to 400 pF (the maximum allowed by I^{2}C) with $R_P = 4.7\,\text{k}\Omega$ (model: $t_r = 1.59\,\mu\text{s}$): which mode still works?

Table 4.3: low level and rise time over R_P ($C_b = 200$ pF)

R_P	$5.9\,\text{k}\Omega$	$4.7\,\text{k}\Omega$	$1.8\,\text{k}\Omega$	$1\,\text{k}\Omega$	$470\,\Omega$	$220\,\Omega$
I_{low}, model	0.56 mA	0.70 mA	1.8 mA	3.2 mA	6.6 mA	13.1 mA
V_{OL}, model	17 mV	22 mV	56 mV	99 mV	204 mV	406 mV
V_{OL}, simulation						
t_r, model	1000 ns	796 ns	305 ns	169 ns	80 ns	37 ns
t_r, simulation						

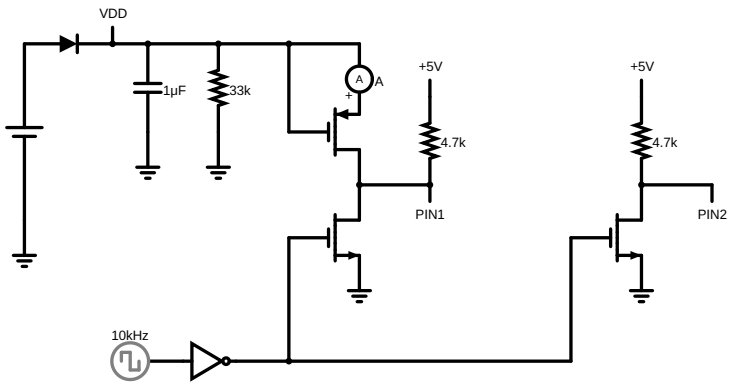
Task 4.5 [S/D] Set the input of device 2 to L. What happens to the clock of device 1? Which I^{2}C feature uses this behaviour? What would happen with two push-pull outputs (part 2)?

Task 4.6 [D] While the line is low, R_P dissipates $P = V_{DD}^2/R_P$ (2.3 mW at $4.7\,\text{k}\Omega$, 11 mW at $1\,\text{k}\Omega$). Since $t_r = 0.8473\,R_P C_b$, the product $P \cdot t_r = 0.8473\,V_{DD}^2 C_b$ does not depend on R_P. What follows for the trade-off between speed and power of an open-drain bus, compared to push-pull? Propose a circuit that speeds up the rising edge without reducing R_P, and try it in the simulation (e.g. a current source pull-up or a switched pull-up).

5 Maximum output voltage: open drain with a 5 V pull-up

▶ open in Falstad (`E_open_drain_overvoltage.txt`)

A 3.3 V microcontroller drives a 5 V logic input through an open-drain output and a pull-up to 5 V. PIN1 is a normal GPIO in open-drain mode: the PMOS is switched off, but it is still there. PIN2 is a 5 V-tolerant open-drain pin without a PMOS. The 3.3 V regulator can only source current (diode). The resistor "MCU load" represents the current consumption of the chip.



Task 5.1 [P/S] Table 5.1 gives the expected high levels and the current that flows from the 5 V supply into the 3.3 V supply via PIN1 ($R_{PU} = 4.7\,\text{k}\Omega$, diode forward voltage about 0.7 V). Explain the values, then measure them.

Table 5.1: open-drain pins with a pull-up to 5 V ($R_{PU} = 4.7\,\text{k}\Omega$)

	high level PIN1	high level PIN2	injection current
model	$V_{DD} + V_F \approx 4.0\,\text{V}$	$5.0\,\text{V}$	$\frac{5 - 4.0}{4.7}\,\text{mA} \approx 0.2\,\text{mA}$
simulation			

Task 5.2 [D] Is the high level of PIN1 sufficient for a 5 V CMOS input with $V_{IH} = 0.7 \cdot 5$ V? Is it allowed?

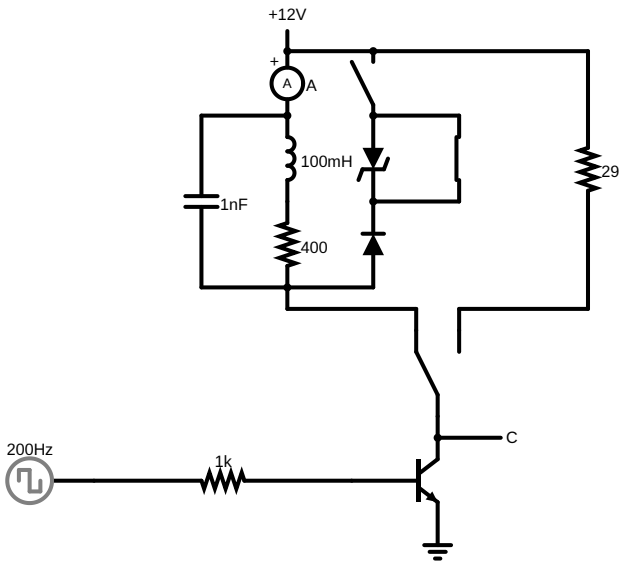
Task 5.3 [S/D] Set $R_{PU} = 1\,\text{k}\Omega$ and keep the MCU load at $33\,\text{k}\Omega$ ($100\,\mu\text{A}$, e.g. in a sleep mode). Observe V_{DD} (simulation: _____) and explain the effect (*back-powering*): the regulator can only source current, so V_{DD} rises until the chip consumes the injected current. What happens if the microcontroller is switched off ($V_{DD} = 0$) but the 5 V side stays on?

Task 5.4 [D] Look up in the data sheets of an RP2040, an RP2350 and an STM32F4 which pins are 5 V tolerant and how large the injection current per pin and in total may be. Name two circuits that shift a 3.3 V open-drain signal to 5 V correctly, one of them bidirectional (for I²C).

6 Open collector: high voltage, high current, inductive loads

► open in Falstad (`F_open_collector_inductive.txt`)

A 3.3 V GPIO switches an NPN transistor ($\beta = 100$) through the base resistor R_B. The collector is an open-collector output for a 12 V load: a relay coil (0.1 H, $400\,\Omega$, 1 nF winding capacitance) or a 12 V/5 W lamp. The switch *flyback* adds a free-wheeling diode, opening the switch *bypass* puts a 24 V Zener diode in series with it. The timescale is shortened (switching at 200 Hz) so that you can see several cycles.



Task 6.1 [P] Table 6.1 gives the operating point with $R_B = 1\,\text{k}\Omega$. Is the transistor saturated?

Table 6.1: operating point, relay coil, $R_B = 1\,\text{k}\Omega$

	$I_C = \frac{12\,\text{V}}{400\,\Omega}$	$I_B = \frac{3.3 - 0.7}{R_B}$	βI_B	overdrive $\frac{\beta I_B}{I_C}$	V_{CE}
model	30 mA	2.6 mA	260 mA	8.7	< 0.2 V
simulation					

Task 6.2 [S/D] Without flyback diode: when the transistor switches off, the coil current continues and charges the winding capacitance. Energy conservation ($\frac{1}{2}LI^2 = \frac{1}{2}CV^2$) gives a peak of $12\,\text{V} + I\sqrt{L/C} = 12\,\text{V} + 300\,\text{V}$. Measure it (simulation: _____) and compare with V_{CEO} of a BC547 (45 V) or of a ULN2003 driver (50 V). What does the simulation not show?

Task 6.3 [S] Close *flyback* and measure the maximum collector voltage and the time until the coil current has fallen to 10 % (right-click the scope, *Properties*, to change the time base). Then open *bypass* (diode + 24 V Zener) and measure again. Explain why the Zener variant is faster. Why does that matter for a relay?

Table 6.2: switching off the relay coil ($\tau = L/R = 0.25\,\text{ms}$)

	diode		diode + 24 V Zener	
	$V_{C,max}$	$t_{10\%}$	$V_{C,max}$	$t_{10\%}$
model	$12 + 0.7 = 12.7$ V	475 µs	$12 + 24 + 0.7 = 36.7$ V	87 µs
simulation				

Model: $i(t) = -V_{cl}/R + (I_0 + V_{cl}/R)\,e^{-t/\tau}$ with the clamp voltage $V_{cl} = V_F$ or $V_Z + V_F$.

Task 6.4 [S/D] Increase R_B to 47 kΩ: $I_B = 55$ µA, so $I_C \le \beta I_B = 5.5$ mA and $V_{CE} \approx 12 - 5.5 \cdot 0.4 = 9.8$ V. Measure I_C and V_{CE} (simulation: _____ and _____). Will the relay still pull in?

Task 6.5 [S/D] Switch the load to the lamp (12 V/5 W, needs 414 mA) and complete Table 6.3. A small-signal transistor in SOT-23 tolerates about 250 mW. Why is 330 Ω still not a good solution? Propose a better driver.

Table 6.3: lamp load

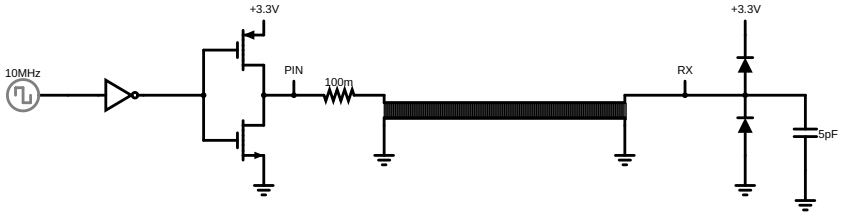
	I_B	I_C	V_{CE}	$P = V_{CE}I_C$
$R_B = 1\,\text{k}\Omega$, model	2.6 mA	$\beta I_B = 260$ mA	$12 - 0.26 \cdot 29 = 4.5$ V	1.2 W
$R_B = 1\,\text{k}\Omega$, simulation				
$R_B = 330\,\Omega$, model	7.9 mA	414 mA (saturated)	≈ 0.1 V	≈ 50 mW
$R_B = 330\,\Omega$, simulation				

Task 6.6 [D] Summarise: what limits the maximum output voltage and the maximum output current of a push-pull pin, of an open-drain pin and of an external open-collector stage?

7 Fast edges on a PCB trace

▶ open in Falstad (`G_transmission_line.txt`)

The push-pull pin drives a $50\,\Omega$ trace with 1 ns delay (about 15 cm on FR4). At the end there is the input of another chip: 5 pF and ESD diodes to V_{DD} and GND. R_S is a series resistor at the driver. The output stage is set to its highest drive strength and fastest slew rate, modelled by $\beta_N = 30$ mA/V^2 and $\beta_P = 25$ mA/V^2. The slew rate is adjustable as on many microcontrollers (RP2040: `SLEWFAST`, STM32: `OSPEEDR`).



Task 7.1 [P] A trace is "electrically long" if the edge is shorter than about twice its delay. Table 7.1 gives the values for $R_S = 0$ and the fastest slew rate. Explain the sequence: launched step, doubling at the open end, clamping, reflection at the driver.

Table 7.1: reflections for $R_S = 0$ ($R_{on,P} = 1/(25\,\text{mA/V}^2 \cdot 2.7\,\text{V}) = 15\,\Omega$)

edge time / round trip	< 1 ns / 2 ns: electrically long
$\Gamma_S = (R_{on} + R_S - Z_0)/(R_{on} + R_S + Z_0)$	$(15 - 50)/(15 + 50) = -0.54$
Γ_L (open end, 5 pF only rounds the edge)	$\approx +1$
launched step $V_{DD}\, Z_0/(Z_0 + R_{on})$	2.54 V
first voltage at RX (before clamping)	$2 \cdot 2.54\,\text{V} = 5.1\,\text{V}$
matched series resistor $R_S = Z_0 - R_{on}$	$\approx 36\,\Omega$ ($33\,\Omega$ standard value)

Task 7.2 [S] Measure overshoot and undershoot at RX and the peak currents in the ESD diodes for the values of R_S in Table 7.2. Which R_S gives the cleanest edge? Compare with $R_S = Z_0 - R_{on}$.

Table 7.2: series termination (fastest slew rate)

R_S	$0\,\Omega$	$10\,\Omega$	$22\,\Omega$	$36\,\Omega$	$50\,\Omega$
$V_{RX,max}$, simulation					
$V_{RX,min}$, simulation					
I_{ESD} to V_{DD} / GND, simulation					

Task 7.3 [S/D] Set R_S back to 0 and reduce the slew rate. From which slew rate on does the ringing disappear? Why should you select the slowest slew rate that meets the timing of your interface?

8 Design task

Task 8.1 [P/S/D] A 3.3 V microcontroller board has to

(a) read a 5 V sensor over I^{2}C at 400 kHz; the cable is 30 cm long, the total bus capacitance is estimated at 250 pF,

(b) switch a 24 V/50 mA solenoid valve at up to 10 Hz,

(c) send a 10 MHz SPI clock over a 10 cm trace to a flash memory.

For each signal choose the output type and configuration of the pin (push-pull/open drain, drive strength, slew rate), dimension the external components and verify your choice in a simulation that you build from the given circuits. Document the remaining risks.